

Soil Water Volume & Water Balance Equations

Generated Summary

June 29, 2025

Notation

- I : Applied irrigation volume (same units as ΔS , e.g. L or gal)
- ΔS : Change in soil water storage (volume)
- Q : Runoff volume
- D : Deep percolation (drainage below measurement depth)
- E : Evaporation and plant uptake losses
- θ_i : Volumetric water content (VWC) at layer i
- Δz_i : Thickness of layer i (soil depth range associated with sensor i)
- A : Cross-sectional area of the strip (width $\times$ unit length)

1. Soil Water Content (SWC)

Given volumetric water content (VWC) measurements θ_d at sensor depths d_1, d_2, d_3 , the weighted average soil water content over the column is

$$\text{SWC} = \sum_{i=1}^3 w_i \theta_{d_i},$$

where w_i are the depth-weights (e.g. $w_1 + w_2 + w_3 = 1$).

2. Water Volume in Soil Column

For a soil column of horizontal cross-sectional area A (e.g. strip width $\times$ length), and volumetric water contents at successive depths, the total water volume V is approximated by:

$$V = A \sum_{i=1}^{n-1} \frac{\theta_{d_i} + \theta_{d_{i+1}}}{2} (z_{i+1} - z_i),$$

where z_i are the depths (e.g. top of reliable range for 6 in sensor to bottom for 18 in).

3. Water Balance

Over an irrigation event:

$$\text{Runoff Volume} = \text{Applied Irrigation Volume} - \Delta S_{\text{soil}},$$

with

$$\Delta S_{\text{soil}} = A \sum_{i=1}^{n-1} \frac{\Delta \theta_{d_i} + \Delta \theta_{d_{i+1}}}{2} (z_{i+1} - z_i).$$

4. Unit Conversions

Using volumetric water content θ (unitless fraction), to convert soil water volume in cubic meters to liters:

$$V_L = V_{\text{m}^3} \times 1000.$$